

Supplementary information for

Wind-Aware, Time-Scale Adaptive PM_{2.5} Spatiotemporal Forecasting with Spatial Regularization

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Table S1. Parameter Explanations for the PM_{2.5} Prediction Model. This table lists the input parameters used in the PM_{2.5} concentration prediction model and provides a brief explanation of their impact on air quality.

Table S2. Air Quality Level Classification Based on the "Technical Regulations for Ambient Air Quality Index"

Fig.S1. Distribution of Stations in the Beijing-Tianjin-Hebei Region.

Fig.S2. Contour maps comparing PM_{2.5} forecasts at different lead times with ground-truth observations. Results are shown for 1h, 6h, 12h, and 24h forecasts across cities in the BTH region under varying pollution levels.

Table S1. Parameter Explanations for PM2.5 Prediction Model

Parameter	Explanation
AQI (Air Quality Index)	A single numerical indicator that integrates multiple pollutant concentrations (e.g., PM _{2.5} , PM ₁₀ , O ₃) to represent overall air quality. It helps track general pollution levels and provides indirect insights into PM _{2.5} trends.
CO (Carbon Monoxide)	Concentration of Carbon Monoxide, mainly from vehicle exhaust and combustion sources. CO often co-occurs with PM _{2.5} in urban areas, indicating shared emission origins.
NO ₂ (Nitrogen Dioxide)	A reactive nitrogen gas produced mainly by fossil fuel combustion. It is a precursor to secondary particulate formation and correlates with traffic-related PM _{2.5} emissions.
O ₃ (Ozone)	A secondary pollutant formed through photochemical reactions involving NO _x and VOCs. Ozone levels influence the oxidative environment and the formation of secondary PM _{2.5} .
PM ₁₀ (Particulate Matter 10)	Particulate matter with diameter ≤10μm, which includes PM _{2.5} . PM10 trends often move with PM2.5 and can indicate broader particle pollution.
PM _{2.5} (Particulate Matter 2.5)	Fine particulate matter with diameter ≤2.5μm. It poses significant health risks and is the core target variable in the prediction model.
SO ₂ (Sulfur Dioxide)	A gaseous pollutant from industrial processes and coal combustion. SO ₂ can react in the atmosphere to form sulfate particles, contributing to PM _{2.5} levels.
t2m (2-meter Temperature)	Near-surface air temperature at 2 meters. It influences atmospheric mixing, stability, and chemical reaction rates, all of which affect PM _{2.5} dynamics.
d2m (2-meter Dew Point Temperature)	Dew point temperature at 2 meters, reflecting ambient humidity. High dew points are associated with enhanced aerosol hygroscopic growth and secondary PM formation.

V10 (10-meter North-South Wind Component)	Meridional (north-south) wind component at 10 meters altitude. It affects pollutant transport and regional dispersion of PM _{2.5} .
u10 (10-meter East-West Wind Component)	Zonal (east-west) wind component at 10 meters altitude. Together with V10, it determines wind-driven horizontal movement of air pollutants.
tp (Total Precipitation Rate)	Total precipitation rate in millimeters per hour. Rainfall helps remove PM _{2.5} from the air through wet deposition.
sp (Surface Pressure)	Surface atmospheric pressure, indicating synoptic weather conditions. It influences boundary layer height and the vertical mixing of pollutants, thus affecting PM _{2.5} accumulation.

Table S2. Air Quality Level Classification Based on the "Technical Regulations for Ambient Air Quality Index"

Level	PM2.5 concentration range (µg/m ³)
Excellent	0-35
Good	36-75
Light Pollution	76-115
Moderate Pollution	116-150
Heavy Pollution	151-250
Severe Pollution	>250

Fig.S1. Distribution of Stations in the Beijing-Tianjin-Hebei Region

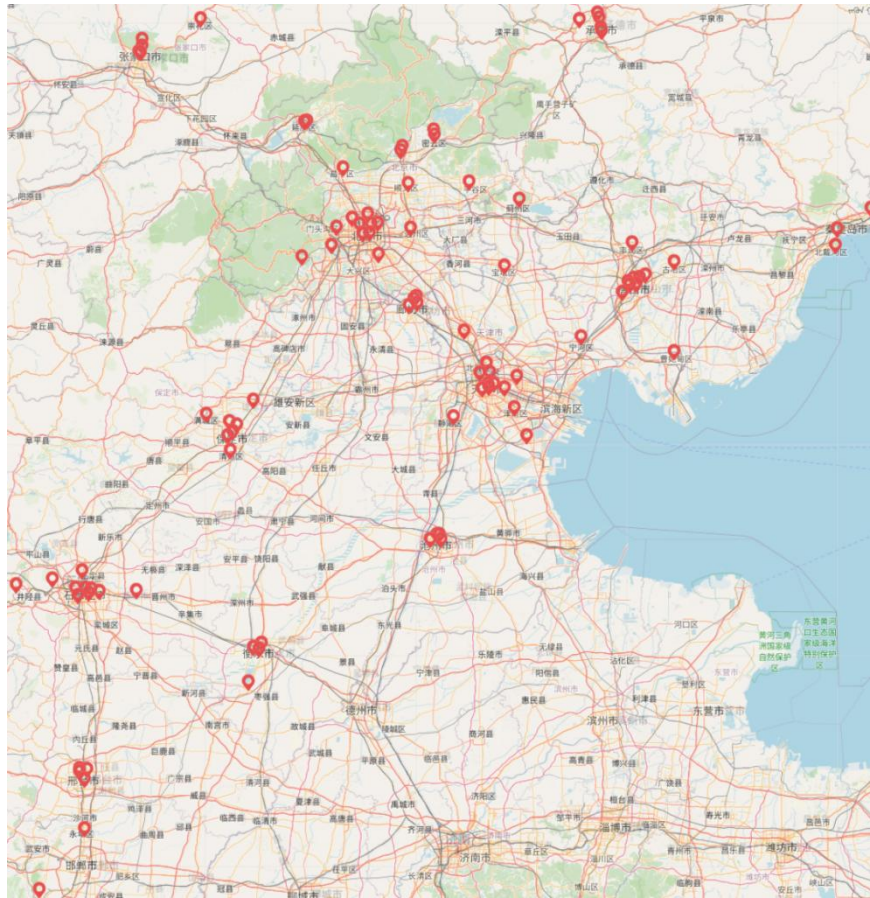


Fig.S2. Contour maps comparing PM_{2.5} forecasts at different lead times with ground-truth observations. Results are shown for 1h, 6h, 12h, and 24h forecasts across cities in the BTH region under varying pollution levels.

